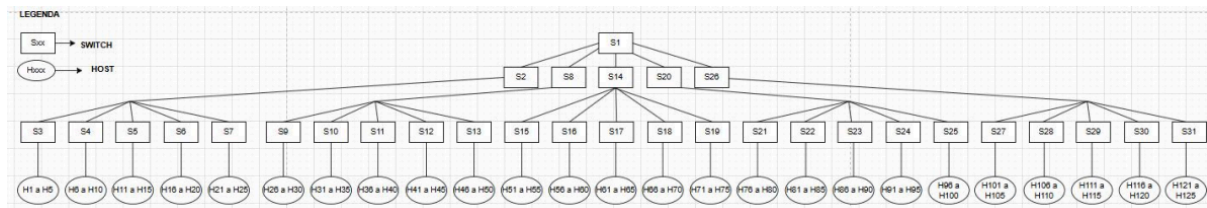


Alunas: Clara de Lima Azevedo e Maria Eduarda de Oliveira Alves

c) topologia



d) testes de ping

```
root@mininet-vm:~# ping -c 5 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data,
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=7.85 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=1.87 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=2.60 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.131 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.112 ms

--- 10.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 0.112/2.515/7.854/2.841 ms
root@mininet-vm:~#
```

```
0x0050: 2627 2829 2a2b 2c2d 2e2f 3031 3233 3435 &'()*+,-./012345
0x0060: 3637
15:48:11.011136 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 27238, seq 5, length
64
0x0000: 0000 0000 0001 0000 0000 0002 0800 4500 .....E.
0x0010: 0054 8a04 0000 4001 dca2 0a00 0002 0a00 .T...@.....
0x0020: 0001 0000 734b 6a66 0005 3be2 0b69 0000 ....skjf...i..
0x0030: 0000 1c2b 0000 0000 0000 1011 1213 1415 ...*+.....%
0x0040: 1617 1819 1a1b 1c1d 1e1f 2021 2223 2425 .....!""#$%
0x0050: 2627 2829 2a2b 2c2d 2e2f 3031 3233 3435 &'()*+,-./012345
0x0060: 3637
15:48:12.018260 ARP, Request who-has 10.0.0.1 tell 10.0.0.2, length 28
0x0000: 0000 0000 0001 0000 0000 0002 0806 0001 .....
0x0010: 0800 0604 0001 0000 0000 0002 0a00 0002 .....
0x0020: 0000 0000 0000 0a00 0001 .....
15:48:12.031187 ARP, Reply 10.0.0.1 is-at 00:00:00:00:00:01, length 28
0x0000: 0000 0000 0002 0000 0000 0001 0806 0001 .....
0x0010: 0800 0604 0002 0000 0000 0001 0a00 0001 .....
0x0020: 0000 0000 0002 0a00 0002 .....
^C
12 packets captured
12 packets received by filter
0 packets dropped by kernel
root@mininet-vm:~#
```

e) testes para larguras de banda bw de 30 Mbps

```
[320] local 10.0.0.1 port 5555 connected with 10.0.0.2 port 60952
[ ID] Interval Transfer Bandwidth
[320] 0.0- 1.0 sec 3.22 MBytes 27.0 Mbits/sec
[320] 1.0- 2.0 sec 3.26 MBytes 27.3 Mbits/sec
[320] 2.0- 3.0 sec 3.29 MBytes 27.6 Mbits/sec
[320] 3.0- 4.0 sec 3.25 MBytes 27.2 Mbits/sec
[320] 4.0- 5.0 sec 3.29 MBytes 27.6 Mbits/sec
[320] 5.0- 6.0 sec 3.32 MBytes 27.8 Mbits/sec
[320] 6.0- 7.0 sec 3.31 MBytes 27.7 Mbits/sec
[320] 7.0- 8.0 sec 3.32 MBytes 27.8 Mbits/sec
[320] 8.0- 9.0 sec 3.32 MBytes 27.8 Mbits/sec
[320] 9.0-10.0 sec 3.33 MBytes 28.0 Mbits/sec
[320] 10.0-11.0 sec 3.30 MBytes 27.7 Mbits/sec
[320] 11.0-12.0 sec 3.33 MBytes 27.9 Mbits/sec
[320] 12.0-13.0 sec 3.31 MBytes 27.8 Mbits/sec
[320] 13.0-14.0 sec 3.34 MBytes 28.0 Mbits/sec
[320] 14.0-15.0 sec 3.32 MBytes 27.9 Mbits/sec
[320] 15.0-16.0 sec 3.34 MBytes 28.0 Mbits/sec
[320] 16.0-17.0 sec 3.33 MBytes 28.0 Mbits/sec
[320] 17.0-18.0 sec 3.36 MBytes 28.2 Mbits/sec
[320] 18.0-19.0 sec 3.36 MBytes 28.1 Mbits/sec
[320] 19.0-20.0 sec 3.32 MBytes 27.8 Mbits/sec
[320] 0.0-20.3 sec 67.1 MBytes 27.8 Mbits/sec
root@mininet-vm:~#
```

```
[319] local 10.0.0.2 port 60952 connected with 10.0.0.1 port 5555
[ ID] Interval Transfer Bandwidth
[319] 0.0- 1.0 sec 3.75 MBytes 31.5 Mbits/sec
[319] 1.0- 2.0 sec 3.12 MBytes 26.2 Mbits/sec
[319] 2.0- 3.0 sec 3.38 MBytes 28.3 Mbits/sec
[319] 3.0- 4.0 sec 3.25 MBytes 27.3 Mbits/sec
[319] 4.0- 5.0 sec 3.38 MBytes 28.3 Mbits/sec
[319] 5.0- 6.0 sec 3.25 MBytes 27.3 Mbits/sec
[319] 6.0- 7.0 sec 3.25 MBytes 27.3 Mbits/sec
[319] 7.0- 8.0 sec 3.50 MBytes 29.4 Mbits/sec
[319] 8.0- 9.0 sec 3.25 MBytes 27.3 Mbits/sec
[319] 9.0-10.0 sec 3.25 MBytes 27.3 Mbits/sec
[319] 10.0-11.0 sec 3.50 MBytes 29.4 Mbits/sec
[319] 11.0-12.0 sec 3.12 MBytes 26.2 Mbits/sec
[319] 12.0-13.0 sec 3.25 MBytes 27.3 Mbits/sec
[319] 13.0-14.0 sec 3.50 MBytes 29.4 Mbits/sec
[319] 14.0-15.0 sec 3.38 MBytes 28.3 Mbits/sec
[319] 15.0-16.0 sec 3.50 MBytes 29.4 Mbits/sec
[319] 16.0-17.0 sec 3.38 MBytes 28.3 Mbits/sec
[319] 17.0-18.0 sec 3.38 MBytes 28.3 Mbits/sec
[319] 18.0-19.0 sec 3.12 MBytes 26.2 Mbits/sec
[319] 19.0-20.0 sec 3.50 MBytes 29.4 Mbits/sec
[319] 0.0-20.0 sec 67.1 MBytes 28.1 Mbits/sec
root@mininet-vm:~#
```

e) testes para larguras de banda bw de 40 Mbps

```

[320] local 10.0.0.1 port 5555 connected with 10.0.0.2 port 32964
[ 11] Interval      Transfer      Bandwidth
[320] 0,0- 1,0 sec  4,35 MBytes  36,5 Mbits/sec
[320] 1,0- 2,0 sec  4,39 MBytes  36,8 Mbits/sec
[320] 2,0- 3,0 sec  4,39 MBytes  36,8 Mbits/sec
[320] 3,0- 4,0 sec  4,31 MBytes  36,2 Mbits/sec
[320] 4,0- 5,0 sec  4,37 MBytes  36,7 Mbits/sec
[320] 5,0- 6,0 sec  4,39 MBytes  36,8 Mbits/sec
[320] 6,0- 7,0 sec  4,41 MBytes  37,0 Mbits/sec
[320] 7,0- 8,0 sec  4,39 MBytes  36,9 Mbits/sec
[320] 8,0- 9,0 sec  4,37 MBytes  36,7 Mbits/sec
[320] 9,0-10,0 sec  4,41 MBytes  37,0 Mbits/sec
[320] 10,0-11,0 sec 4,43 MBytes  37,2 Mbits/sec
[320] 11,0-12,0 sec 4,39 MBytes  36,8 Mbits/sec
[320] 12,0-13,0 sec 4,39 MBytes  36,8 Mbits/sec
[320] 13,0-14,0 sec 4,39 MBytes  36,9 Mbits/sec
[320] 14,0-15,0 sec 4,42 MBytes  37,1 Mbits/sec
[320] 15,0-16,0 sec 4,41 MBytes  37,0 Mbits/sec
[320] 16,0-17,0 sec 4,48 MBytes  37,5 Mbits/sec
[320] 17,0-18,0 sec 4,45 MBytes  37,3 Mbits/sec
[320] 18,0-19,0 sec 4,47 MBytes  37,5 Mbits/sec
[320] 19,0-20,0 sec 4,44 MBytes  37,3 Mbits/sec
[320] 0,0-20,2 sec 89,0 MBytes  36,9 Mbits/sec

[319] local 10.0.0.2 port 32964 connected with 10.0.0.1 port 5555
[ 11] Interval      Transfer      Bandwidth
[319] 0,0- 1,0 sec  4,88 MBytes  40,9 Mbits/sec
[319] 1,0- 2,0 sec  4,38 MBytes  36,7 Mbits/sec
[319] 2,0- 3,0 sec  4,50 MBytes  37,7 Mbits/sec
[319] 3,0- 4,0 sec  4,25 MBytes  35,7 Mbits/sec
[319] 4,0- 5,0 sec  4,50 MBytes  37,7 Mbits/sec
[319] 5,0- 6,0 sec  4,25 MBytes  35,7 Mbits/sec
[319] 6,0- 7,0 sec  4,38 MBytes  36,7 Mbits/sec
[319] 7,0- 8,0 sec  4,50 MBytes  37,7 Mbits/sec
[319] 8,0- 9,0 sec  4,25 MBytes  35,7 Mbits/sec
[319] 9,0-10,0 sec 4,50 MBytes  37,7 Mbits/sec
[319] 10,0-11,0 sec 4,50 MBytes  37,7 Mbits/sec
[319] 11,0-12,0 sec 4,38 MBytes  36,7 Mbits/sec
[319] 12,0-13,0 sec 4,25 MBytes  35,7 Mbits/sec
[319] 13,0-14,0 sec 4,50 MBytes  37,7 Mbits/sec
[319] 14,0-15,0 sec 4,88 MBytes  40,9 Mbits/sec
[319] 15,0-16,0 sec 4,12 MBytes  34,6 Mbits/sec
[319] 16,0-17,0 sec 4,50 MBytes  37,7 Mbits/sec
[319] 17,0-18,0 sec 4,50 MBytes  37,7 Mbits/sec
[319] 18,0-19,0 sec 4,38 MBytes  36,7 Mbits/sec
[319] 19,0-20,0 sec 4,50 MBytes  37,7 Mbits/sec
[319] 0,0-20,0 sec 89,0 MBytes  37,3 Mbits/sec

root@mininet-vm:~#

```

a) b) linhas de comando

- Criação da topologia:
`sudo mn -c`
`sudo mn --topo tree,depth=3,fanout=5 --mac --link tc,bw=30`

Explicação:

`sudo mn -c` → remove redes anteriores
`--topo tree,profundidade=3,fanout=5` → cria uma topologia em árvore com 3 níveis e 5 ramos por switch.
`--mac` → define endereços MAC automáticos e padronizados.
`--link tc,bw=30` → define a largura da banda dos links como 30 Mbps.

- Visualizar os nós, conexões e interfaces
`nodes`
`net`
`dump`

Explicação:

`nodes` → lista hosts e switches
`net` → mostra ligações entre os nós
`dump` → mostra IPs, MACs e portas

mininet@mininet-vm:~\$ `sudo mn -c`

```

*** Removing excess controllers/ofprotocols/ofdatapaths/pings/noxes
killall controller ofprotocol ofdatapath ping nox_core lt-nox_core ovs-openflowd ovs-controller
udpbwtest mnexec ivs 2> /dev/null
killall -9 controller ofprotocol ofdatapath ping nox_core lt-nox_core ovs-openflowd
ovs-controller udpbwtest mnexec ivs 2> /dev/null
pkill -9 -f "sudo mnexec"
*** Removing junk from /tmp
rm -f /tmp/vconn* /tmp/vlogs* /tmp/*.out /tmp/*.log
*** Removing old X11 tunnels

```

```

*** Removing excess kernel datapaths
ps ax | egrep -o 'dp[0-9]+' | sed 's/dp/nl:/'
*** Removing OVS datapaths
ovs-vsctl --timeout=1 list-br
ovs-vsctl --timeout=1 list-br
*** Removing all links of the pattern foo-ethX
ip link show | egrep -o '([_[:alnum:]]+-eth[[:digit:]]+)'
ip link show
*** Killing stale mininet node processes
pkill -9 -f mininet:
*** Shutting down stale tunnels
pkill -9 -f Tunnel=Ethernet
pkill -9 -f .ssh/mn
rm -f ~/.ssh/mn/*
*** Cleanup complete.
mininet@mininet-vm:~$ sudo mn --topo=tree,depth=3,fanout=5 --mac --link=tc,bw=30
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h21 h22 h23
h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h41 h42 h43 h44
h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h61 h62 h63 h64 h65
h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h82 h83 h84 h85 h86
h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101 h102 h103 h104 h105
h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117 h118 h119 h120 h121
h122 h123 h124 h125
*** Adding switches:
s1 s2 s3 s4 s5 s6 s7 s8 s9 s10 s11 s12 s13 s14 s15 s16 s17 s18 s19 s20 s21 s22 s23 s24
s25 s26 s27 s28 s29 s30 s31
*** Adding links:
(30.00Mbit) (30.00Mbit) (s1, s2) (30.00Mbit) (30.00Mbit) (s1, s8) (30.00Mbit) (30.00Mbit) (s1,
s14) (30.00Mbit) (30.00Mbit) (s1, s20) (30.00Mbit) (30.00Mbit) (s1, s26) (30.00Mbit)
(30.00Mbit) (s2, s3) (30.00Mbit) (30.00Mbit) (s2, s4) (30.00Mbit) (30.00Mbit) (s2, s5)
(30.00Mbit) (30.00Mbit) (s2, s6) (30.00Mbit) (30.00Mbit) (s2, s7) (30.00Mbit) (30.00Mbit) (s3,
h1) (30.00Mbit) (30.00Mbit) (s3, h2) (30.00Mbit) (30.00Mbit) (s3, h3) (30.00Mbit) (30.00Mbit)
(s3, h4) (30.00Mbit) (30.00Mbit) (s3, h5) (30.00Mbit) (30.00Mbit) (s4, h6) (30.00Mbit)
(30.00Mbit) (s4, h7) (30.00Mbit) (30.00Mbit) (s4, h8) (30.00Mbit) (30.00Mbit) (s4, h9)
(30.00Mbit) (30.00Mbit) (s4, h10) (30.00Mbit) (30.00Mbit) (s5, h11) (30.00Mbit) (30.00Mbit)
(s5, h12) (30.00Mbit) (30.00Mbit) (s5, h13) (30.00Mbit) (30.00Mbit) (s5, h14) (30.00Mbit)
(30.00Mbit) (s5, h15) (30.00Mbit) (30.00Mbit) (s6, h16) (30.00Mbit) (30.00Mbit) (s6, h17)
(30.00Mbit) (30.00Mbit) (s6, h18) (30.00Mbit) (30.00Mbit) (s6, h19) (30.00Mbit) (30.00Mbit)
(s6, h20) (30.00Mbit) (30.00Mbit) (s7, h21) (30.00Mbit) (30.00Mbit) (s7, h22) (30.00Mbit)
(30.00Mbit) (s7, h23) (30.00Mbit) (30.00Mbit) (s7, h24) (30.00Mbit) (30.00Mbit) (s7, h25)
(30.00Mbit) (30.00Mbit) (s8, s9) (30.00Mbit) (30.00Mbit) (s8, s10) (30.00Mbit) (30.00Mbit)
(s8, s11) (30.00Mbit) (30.00Mbit) (s8, s12) (30.00Mbit) (30.00Mbit) (s8, s13) (30.00Mbit)
(30.00Mbit) (s9, h26) (30.00Mbit) (30.00Mbit) (s9, h27) (30.00Mbit) (30.00Mbit) (s9, h28)
(30.00Mbit) (30.00Mbit) (s9, h29) (30.00Mbit) (30.00Mbit) (s9, h30) (30.00Mbit) (30.00Mbit)
(s10, h31) (30.00Mbit) (30.00Mbit) (s10, h32) (30.00Mbit) (30.00Mbit) (s10, h33) (30.00Mbit)

```

[illegible]

h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h21 h22 h23
h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h41 h42 h43 h44
h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h61 h62 h63 h64 h65
h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h82 h83 h84 h85 h86

h3 h3-eth0:s3-eth3
h4 h4-eth0:s3-eth4
h5 h5-eth0:s3-eth5
h6 h6-eth0:s4-eth1
h7 h7-eth0:s4-eth2
h8 h8-eth0:s4-eth3
h9 h9-eth0:s4-eth4
h10 h10-eth0:s4-eth5
h11 h11-eth0:s5-eth1
h12 h12-eth0:s5-eth2
h13 h13-eth0:s5-eth3
h14 h14-eth0:s5-eth4
h15 h15-eth0:s5-eth5
h16 h16-eth0:s6-eth1
h17 h17-eth0:s6-eth2
h18 h18-eth0:s6-eth3
h19 h19-eth0:s6-eth4
h20 h20-eth0:s6-eth5
h21 h21-eth0:s7-eth1
h22 h22-eth0:s7-eth2
h23 h23-eth0:s7-eth3
h24 h24-eth0:s7-eth4
h25 h25-eth0:s7-eth5
h26 h26-eth0:s9-eth1
h27 h27-eth0:s9-eth2
h28 h28-eth0:s9-eth3
h29 h29-eth0:s9-eth4
h30 h30-eth0:s9-eth5
h31 h31-eth0:s10-eth1
h32 h32-eth0:s10-eth2
h33 h33-eth0:s10-eth3
h34 h34-eth0:s10-eth4
h35 h35-eth0:s10-eth5
h36 h36-eth0:s11-eth1
h37 h37-eth0:s11-eth2
h38 h38-eth0:s11-eth3
h39 h39-eth0:s11-eth4
h40 h40-eth0:s11-eth5
h41 h41-eth0:s12-eth1
h42 h42-eth0:s12-eth2
h43 h43-eth0:s12-eth3
h44 h44-eth0:s12-eth4
h45 h45-eth0:s12-eth5
h46 h46-eth0:s13-eth1
h47 h47-eth0:s13-eth2
h48 h48-eth0:s13-eth3
h49 h49-eth0:s13-eth4
h50 h50-eth0:s13-eth5

h51 h51-eth0:s15-eth1
h52 h52-eth0:s15-eth2
h53 h53-eth0:s15-eth3
h54 h54-eth0:s15-eth4
h55 h55-eth0:s15-eth5
h56 h56-eth0:s16-eth1
h57 h57-eth0:s16-eth2
h58 h58-eth0:s16-eth3
h59 h59-eth0:s16-eth4
h60 h60-eth0:s16-eth5
h61 h61-eth0:s17-eth1
h62 h62-eth0:s17-eth2
h63 h63-eth0:s17-eth3
h64 h64-eth0:s17-eth4
h65 h65-eth0:s17-eth5
h66 h66-eth0:s18-eth1
h67 h67-eth0:s18-eth2
h68 h68-eth0:s18-eth3
h69 h69-eth0:s18-eth4
h70 h70-eth0:s18-eth5
h71 h71-eth0:s19-eth1
h72 h72-eth0:s19-eth2
h73 h73-eth0:s19-eth3
h74 h74-eth0:s19-eth4
h75 h75-eth0:s19-eth5
h76 h76-eth0:s21-eth1
h77 h77-eth0:s21-eth2
h78 h78-eth0:s21-eth3
h79 h79-eth0:s21-eth4
h80 h80-eth0:s21-eth5
h81 h81-eth0:s22-eth1
h82 h82-eth0:s22-eth2
h83 h83-eth0:s22-eth3
h84 h84-eth0:s22-eth4
h85 h85-eth0:s22-eth5
h86 h86-eth0:s23-eth1
h87 h87-eth0:s23-eth2
h88 h88-eth0:s23-eth3
h89 h89-eth0:s23-eth4
h90 h90-eth0:s23-eth5
h91 h91-eth0:s24-eth1
h92 h92-eth0:s24-eth2
h93 h93-eth0:s24-eth3
h94 h94-eth0:s24-eth4
h95 h95-eth0:s24-eth5
h96 h96-eth0:s25-eth1
h97 h97-eth0:s25-eth2
h98 h98-eth0:s25-eth3

h99 h99-eth0:s25-eth4
h100 h100-eth0:s25-eth5
h101 h101-eth0:s27-eth1
h102 h102-eth0:s27-eth2
h103 h103-eth0:s27-eth3
h104 h104-eth0:s27-eth4
h105 h105-eth0:s27-eth5
h106 h106-eth0:s28-eth1
h107 h107-eth0:s28-eth2
h108 h108-eth0:s28-eth3
h109 h109-eth0:s28-eth4
h110 h110-eth0:s28-eth5
h111 h111-eth0:s29-eth1
h112 h112-eth0:s29-eth2
h113 h113-eth0:s29-eth3
h114 h114-eth0:s29-eth4
h115 h115-eth0:s29-eth5
h116 h116-eth0:s30-eth1
h117 h117-eth0:s30-eth2
h118 h118-eth0:s30-eth3
h119 h119-eth0:s30-eth4
h120 h120-eth0:s30-eth5
h121 h121-eth0:s31-eth1
h122 h122-eth0:s31-eth2
h123 h123-eth0:s31-eth3
h124 h124-eth0:s31-eth4
h125 h125-eth0:s31-eth5
s1 lo: s1-eth1:s2-eth6 s1-eth2:s8-eth6 s1-eth3:s14-eth6 s1-eth4:s20-eth6 s1-eth5:s26-eth6
s2 lo: s2-eth1:s3-eth6 s2-eth2:s4-eth6 s2-eth3:s5-eth6 s2-eth4:s6-eth6 s2-eth5:s7-eth6
s2-eth6:s1-eth1
s3 lo: s3-eth1:h1-eth0 s3-eth2:h2-eth0 s3-eth3:h3-eth0 s3-eth4:h4-eth0 s3-eth5:h5-eth0
s3-eth6:s2-eth1
s4 lo: s4-eth1:h6-eth0 s4-eth2:h7-eth0 s4-eth3:h8-eth0 s4-eth4:h9-eth0 s4-eth5:h10-eth0
s4-eth6:s2-eth2
s5 lo: s5-eth1:h11-eth0 s5-eth2:h12-eth0 s5-eth3:h13-eth0 s5-eth4:h14-eth0
s5-eth5:h15-eth0 s5-eth6:s2-eth3
s6 lo: s6-eth1:h16-eth0 s6-eth2:h17-eth0 s6-eth3:h18-eth0 s6-eth4:h19-eth0
s6-eth5:h20-eth0 s6-eth6:s2-eth4
s7 lo: s7-eth1:h21-eth0 s7-eth2:h22-eth0 s7-eth3:h23-eth0 s7-eth4:h24-eth0
s7-eth5:h25-eth0 s7-eth6:s2-eth5
s8 lo: s8-eth1:s9-eth6 s8-eth2:s10-eth6 s8-eth3:s11-eth6 s8-eth4:s12-eth6 s8-eth5:s13-eth6
s8-eth6:s1-eth2
s9 lo: s9-eth1:h26-eth0 s9-eth2:h27-eth0 s9-eth3:h28-eth0 s9-eth4:h29-eth0
s9-eth5:h30-eth0 s9-eth6:s8-eth1
s10 lo: s10-eth1:h31-eth0 s10-eth2:h32-eth0 s10-eth3:h33-eth0 s10-eth4:h34-eth0
s10-eth5:h35-eth0 s10-eth6:s8-eth2
s11 lo: s11-eth1:h36-eth0 s11-eth2:h37-eth0 s11-eth3:h38-eth0 s11-eth4:h39-eth0
s11-eth5:h40-eth0 s11-eth6:s8-eth3

s12 lo: s12-eth1:h41-eth0 s12-eth2:h42-eth0 s12-eth3:h43-eth0 s12-eth4:h44-eth0
s12-eth5:h45-eth0 s12-eth6:s8-eth4
s13 lo: s13-eth1:h46-eth0 s13-eth2:h47-eth0 s13-eth3:h48-eth0 s13-eth4:h49-eth0
s13-eth5:h50-eth0 s13-eth6:s8-eth5
s14 lo: s14-eth1:s15-eth6 s14-eth2:s16-eth6 s14-eth3:s17-eth6 s14-eth4:s18-eth6
s14-eth5:s19-eth6 s14-eth6:s1-eth3
s15 lo: s15-eth1:h51-eth0 s15-eth2:h52-eth0 s15-eth3:h53-eth0 s15-eth4:h54-eth0
s15-eth5:h55-eth0 s15-eth6:s14-eth1
s16 lo: s16-eth1:h56-eth0 s16-eth2:h57-eth0 s16-eth3:h58-eth0 s16-eth4:h59-eth0
s16-eth5:h60-eth0 s16-eth6:s14-eth2
s17 lo: s17-eth1:h61-eth0 s17-eth2:h62-eth0 s17-eth3:h63-eth0 s17-eth4:h64-eth0
s17-eth5:h65-eth0 s17-eth6:s14-eth3
s18 lo: s18-eth1:h66-eth0 s18-eth2:h67-eth0 s18-eth3:h68-eth0 s18-eth4:h69-eth0
s18-eth5:h70-eth0 s18-eth6:s14-eth4
s19 lo: s19-eth1:h71-eth0 s19-eth2:h72-eth0 s19-eth3:h73-eth0 s19-eth4:h74-eth0
s19-eth5:h75-eth0 s19-eth6:s14-eth5
s20 lo: s20-eth1:s21-eth6 s20-eth2:s22-eth6 s20-eth3:s23-eth6 s20-eth4:s24-eth6
s20-eth5:s25-eth6 s20-eth6:s1-eth4
s21 lo: s21-eth1:h76-eth0 s21-eth2:h77-eth0 s21-eth3:h78-eth0 s21-eth4:h79-eth0
s21-eth5:h80-eth0 s21-eth6:s20-eth1
s22 lo: s22-eth1:h81-eth0 s22-eth2:h82-eth0 s22-eth3:h83-eth0 s22-eth4:h84-eth0
s22-eth5:h85-eth0 s22-eth6:s20-eth2
s23 lo: s23-eth1:h86-eth0 s23-eth2:h87-eth0 s23-eth3:h88-eth0 s23-eth4:h89-eth0
s23-eth5:h90-eth0 s23-eth6:s20-eth3
s24 lo: s24-eth1:h91-eth0 s24-eth2:h92-eth0 s24-eth3:h93-eth0 s24-eth4:h94-eth0
s24-eth5:h95-eth0 s24-eth6:s20-eth4
s25 lo: s25-eth1:h96-eth0 s25-eth2:h97-eth0 s25-eth3:h98-eth0 s25-eth4:h99-eth0
s25-eth5:h100-eth0 s25-eth6:s20-eth5
s26 lo: s26-eth1:s27-eth6 s26-eth2:s28-eth6 s26-eth3:s29-eth6 s26-eth4:s30-eth6
s26-eth5:s31-eth6 s26-eth6:s1-eth5
s27 lo: s27-eth1:h101-eth0 s27-eth2:h102-eth0 s27-eth3:h103-eth0 s27-eth4:h104-eth0
s27-eth5:h105-eth0 s27-eth6:s26-eth1
s28 lo: s28-eth1:h106-eth0 s28-eth2:h107-eth0 s28-eth3:h108-eth0 s28-eth4:h109-eth0
s28-eth5:h110-eth0 s28-eth6:s26-eth2
s29 lo: s29-eth1:h111-eth0 s29-eth2:h112-eth0 s29-eth3:h113-eth0 s29-eth4:h114-eth0
s29-eth5:h115-eth0 s29-eth6:s26-eth3
s30 lo: s30-eth1:h116-eth0 s30-eth2:h117-eth0 s30-eth3:h118-eth0 s30-eth4:h119-eth0
s30-eth5:h120-eth0 s30-eth6:s26-eth4
s31 lo: s31-eth1:h121-eth0 s31-eth2:h122-eth0 s31-eth3:h123-eth0 s31-eth4:h124-eth0
s31-eth5:h125-eth0 s31-eth6:s26-eth5

c0

mininet> dump

<Host h1: h1-eth0:10.0.0.1 pid=12400>
<Host h2: h2-eth0:10.0.0.2 pid=12402>
<Host h3: h3-eth0:10.0.0.3 pid=12404>
<Host h4: h4-eth0:10.0.0.4 pid=12406>
<Host h5: h5-eth0:10.0.0.5 pid=12408>
<Host h6: h6-eth0:10.0.0.6 pid=12410>

<Host h7: h7-eth0:10.0.0.7 pid=12412>
<Host h8: h8-eth0:10.0.0.8 pid=12414>
<Host h9: h9-eth0:10.0.0.9 pid=12416>
<Host h10: h10-eth0:10.0.0.10 pid=12418>
<Host h11: h11-eth0:10.0.0.11 pid=12420>
<Host h12: h12-eth0:10.0.0.12 pid=12422>
<Host h13: h13-eth0:10.0.0.13 pid=12424>
<Host h14: h14-eth0:10.0.0.14 pid=12426>
<Host h15: h15-eth0:10.0.0.15 pid=12428>
<Host h16: h16-eth0:10.0.0.16 pid=12430>
<Host h17: h17-eth0:10.0.0.17 pid=12432>
<Host h18: h18-eth0:10.0.0.18 pid=12434>
<Host h19: h19-eth0:10.0.0.19 pid=12436>
<Host h20: h20-eth0:10.0.0.20 pid=12438>
<Host h21: h21-eth0:10.0.0.21 pid=12440>
<Host h22: h22-eth0:10.0.0.22 pid=12442>
<Host h23: h23-eth0:10.0.0.23 pid=12444>
<Host h24: h24-eth0:10.0.0.24 pid=12446>
<Host h25: h25-eth0:10.0.0.25 pid=12448>
<Host h26: h26-eth0:10.0.0.26 pid=12450>
<Host h27: h27-eth0:10.0.0.27 pid=12452>
<Host h28: h28-eth0:10.0.0.28 pid=12454>
<Host h29: h29-eth0:10.0.0.29 pid=12456>
<Host h30: h30-eth0:10.0.0.30 pid=12458>
<Host h31: h31-eth0:10.0.0.31 pid=12460>
<Host h32: h32-eth0:10.0.0.32 pid=12462>
<Host h33: h33-eth0:10.0.0.33 pid=12464>
<Host h34: h34-eth0:10.0.0.34 pid=12466>
<Host h35: h35-eth0:10.0.0.35 pid=12468>
<Host h36: h36-eth0:10.0.0.36 pid=12470>
<Host h37: h37-eth0:10.0.0.37 pid=12472>
<Host h38: h38-eth0:10.0.0.38 pid=12474>
<Host h39: h39-eth0:10.0.0.39 pid=12476>
<Host h40: h40-eth0:10.0.0.40 pid=12478>
<Host h41: h41-eth0:10.0.0.41 pid=12480>
<Host h42: h42-eth0:10.0.0.42 pid=12482>
<Host h43: h43-eth0:10.0.0.43 pid=12484>
<Host h44: h44-eth0:10.0.0.44 pid=12486>
<Host h45: h45-eth0:10.0.0.45 pid=12488>
<Host h46: h46-eth0:10.0.0.46 pid=12490>
<Host h47: h47-eth0:10.0.0.47 pid=12492>
<Host h48: h48-eth0:10.0.0.48 pid=12494>
<Host h49: h49-eth0:10.0.0.49 pid=12496>
<Host h50: h50-eth0:10.0.0.50 pid=12498>
<Host h51: h51-eth0:10.0.0.51 pid=12502>
<Host h52: h52-eth0:10.0.0.52 pid=12504>
<Host h53: h53-eth0:10.0.0.53 pid=12506>
<Host h54: h54-eth0:10.0.0.54 pid=12508>

<Host h55: h55-eth0:10.0.0.55 pid=12510>
<Host h56: h56-eth0:10.0.0.56 pid=12512>
<Host h57: h57-eth0:10.0.0.57 pid=12514>
<Host h58: h58-eth0:10.0.0.58 pid=12516>
<Host h59: h59-eth0:10.0.0.59 pid=12518>
<Host h60: h60-eth0:10.0.0.60 pid=12520>
<Host h61: h61-eth0:10.0.0.61 pid=12522>
<Host h62: h62-eth0:10.0.0.62 pid=12524>
<Host h63: h63-eth0:10.0.0.63 pid=12526>
<Host h64: h64-eth0:10.0.0.64 pid=12528>
<Host h65: h65-eth0:10.0.0.65 pid=12530>
<Host h66: h66-eth0:10.0.0.66 pid=12532>
<Host h67: h67-eth0:10.0.0.67 pid=12534>
<Host h68: h68-eth0:10.0.0.68 pid=12536>
<Host h69: h69-eth0:10.0.0.69 pid=12538>
<Host h70: h70-eth0:10.0.0.70 pid=12540>
<Host h71: h71-eth0:10.0.0.71 pid=12542>
<Host h72: h72-eth0:10.0.0.72 pid=12544>
<Host h73: h73-eth0:10.0.0.73 pid=12546>
<Host h74: h74-eth0:10.0.0.74 pid=12548>
<Host h75: h75-eth0:10.0.0.75 pid=12550>
<Host h76: h76-eth0:10.0.0.76 pid=12552>
<Host h77: h77-eth0:10.0.0.77 pid=12554>
<Host h78: h78-eth0:10.0.0.78 pid=12556>
<Host h79: h79-eth0:10.0.0.79 pid=12558>
<Host h80: h80-eth0:10.0.0.80 pid=12560>
<Host h81: h81-eth0:10.0.0.81 pid=12562>
<Host h82: h82-eth0:10.0.0.82 pid=12564>
<Host h83: h83-eth0:10.0.0.83 pid=12566>
<Host h84: h84-eth0:10.0.0.84 pid=12568>
<Host h85: h85-eth0:10.0.0.85 pid=12570>
<Host h86: h86-eth0:10.0.0.86 pid=12572>
<Host h87: h87-eth0:10.0.0.87 pid=12574>
<Host h88: h88-eth0:10.0.0.88 pid=12576>
<Host h89: h89-eth0:10.0.0.89 pid=12578>
<Host h90: h90-eth0:10.0.0.90 pid=12580>
<Host h91: h91-eth0:10.0.0.91 pid=12582>
<Host h92: h92-eth0:10.0.0.92 pid=12584>
<Host h93: h93-eth0:10.0.0.93 pid=12586>
<Host h94: h94-eth0:10.0.0.94 pid=12588>
<Host h95: h95-eth0:10.0.0.95 pid=12590>
<Host h96: h96-eth0:10.0.0.96 pid=12592>
<Host h97: h97-eth0:10.0.0.97 pid=12594>
<Host h98: h98-eth0:10.0.0.98 pid=12596>
<Host h99: h99-eth0:10.0.0.99 pid=12598>
<Host h100: h100-eth0:10.0.0.100 pid=12600>
<Host h101: h101-eth0:10.0.0.101 pid=12602>
<Host h102: h102-eth0:10.0.0.102 pid=12604>

<Host h103: h103-eth0:10.0.0.103 pid=12606>
<Host h104: h104-eth0:10.0.0.104 pid=12608>
<Host h105: h105-eth0:10.0.0.105 pid=12610>
<Host h106: h106-eth0:10.0.0.106 pid=12612>
<Host h107: h107-eth0:10.0.0.107 pid=12614>
<Host h108: h108-eth0:10.0.0.108 pid=12616>
<Host h109: h109-eth0:10.0.0.109 pid=12618>
<Host h110: h110-eth0:10.0.0.110 pid=12620>
<Host h111: h111-eth0:10.0.0.111 pid=12622>
<Host h112: h112-eth0:10.0.0.112 pid=12624>
<Host h113: h113-eth0:10.0.0.113 pid=12626>
<Host h114: h114-eth0:10.0.0.114 pid=12628>
<Host h115: h115-eth0:10.0.0.115 pid=12630>
<Host h116: h116-eth0:10.0.0.116 pid=12632>
<Host h117: h117-eth0:10.0.0.117 pid=12634>
<Host h118: h118-eth0:10.0.0.118 pid=12636>
<Host h119: h119-eth0:10.0.0.119 pid=12638>
<Host h120: h120-eth0:10.0.0.120 pid=12640>
<Host h121: h121-eth0:10.0.0.121 pid=12642>
<Host h122: h122-eth0:10.0.0.122 pid=12644>
<Host h123: h123-eth0:10.0.0.123 pid=12646>
<Host h124: h124-eth0:10.0.0.124 pid=12648>
<Host h125: h125-eth0:10.0.0.125 pid=12650>
<OVSSwitch s1:
lo:127.0.0.1,s1-eth1:None,s1-eth2:None,s1-eth3:None,s1-eth4:None,s1-eth5:None
pid=12655>
<OVSSwitch s2:
lo:127.0.0.1,s2-eth1:None,s2-eth2:None,s2-eth3:None,s2-eth4:None,s2-eth5:None,s2-eth6:
None pid=12658>
<OVSSwitch s3:
lo:127.0.0.1,s3-eth1:None,s3-eth2:None,s3-eth3:None,s3-eth4:None,s3-eth5:None,s3-eth6:
None pid=12661>
<OVSSwitch s4:
lo:127.0.0.1,s4-eth1:None,s4-eth2:None,s4-eth3:None,s4-eth4:None,s4-eth5:None,s4-eth6:
None pid=12664>
<OVSSwitch s5:
lo:127.0.0.1,s5-eth1:None,s5-eth2:None,s5-eth3:None,s5-eth4:None,s5-eth5:None,s5-eth6:
None pid=12667>
<OVSSwitch s6:
lo:127.0.0.1,s6-eth1:None,s6-eth2:None,s6-eth3:None,s6-eth4:None,s6-eth5:None,s6-eth6:
None pid=12670>
<OVSSwitch s7:
lo:127.0.0.1,s7-eth1:None,s7-eth2:None,s7-eth3:None,s7-eth4:None,s7-eth5:None,s7-eth6:
None pid=12673>
<OVSSwitch s8:
lo:127.0.0.1,s8-eth1:None,s8-eth2:None,s8-eth3:None,s8-eth4:None,s8-eth5:None,s8-eth6:
None pid=12676>

<OVSSwitch s9:
lo:127.0.0.1,s9-eth1:None,s9-eth2:None,s9-eth3:None,s9-eth4:None,s9-eth5:None,s9-eth6:
None pid=12679>
<OVSSwitch s10:
lo:127.0.0.1,s10-eth1:None,s10-eth2:None,s10-eth3:None,s10-eth4:None,s10-eth5:None,s1
0-eth6:None pid=12682>
<OVSSwitch s11:
lo:127.0.0.1,s11-eth1:None,s11-eth2:None,s11-eth3:None,s11-eth4:None,s11-eth5:None,s11
-eth6:None pid=12685>
<OVSSwitch s12:
lo:127.0.0.1,s12-eth1:None,s12-eth2:None,s12-eth3:None,s12-eth4:None,s12-eth5:None,s1
2-eth6:None pid=12688>
<OVSSwitch s13:
lo:127.0.0.1,s13-eth1:None,s13-eth2:None,s13-eth3:None,s13-eth4:None,s13-eth5:None,s1
3-eth6:None pid=12691>
<OVSSwitch s14:
lo:127.0.0.1,s14-eth1:None,s14-eth2:None,s14-eth3:None,s14-eth4:None,s14-eth5:None,s1
4-eth6:None pid=12694>
<OVSSwitch s15:
lo:127.0.0.1,s15-eth1:None,s15-eth2:None,s15-eth3:None,s15-eth4:None,s15-eth5:None,s1
5-eth6:None pid=12697>
<OVSSwitch s16:
lo:127.0.0.1,s16-eth1:None,s16-eth2:None,s16-eth3:None,s16-eth4:None,s16-eth5:None,s1
6-eth6:None pid=12700>
<OVSSwitch s17:
lo:127.0.0.1,s17-eth1:None,s17-eth2:None,s17-eth3:None,s17-eth4:None,s17-eth5:None,s1
7-eth6:None pid=12703>
<OVSSwitch s18:
lo:127.0.0.1,s18-eth1:None,s18-eth2:None,s18-eth3:None,s18-eth4:None,s18-eth5:None,s1
8-eth6:None pid=12706>
<OVSSwitch s19:
lo:127.0.0.1,s19-eth1:None,s19-eth2:None,s19-eth3:None,s19-eth4:None,s19-eth5:None,s1
9-eth6:None pid=12709>
<OVSSwitch s20:
lo:127.0.0.1,s20-eth1:None,s20-eth2:None,s20-eth3:None,s20-eth4:None,s20-eth5:None,s2
0-eth6:None pid=12712>
<OVSSwitch s21:
lo:127.0.0.1,s21-eth1:None,s21-eth2:None,s21-eth3:None,s21-eth4:None,s21-eth5:None,s2
1-eth6:None pid=12715>
<OVSSwitch s22:
lo:127.0.0.1,s22-eth1:None,s22-eth2:None,s22-eth3:None,s22-eth4:None,s22-eth5:None,s2
2-eth6:None pid=12718>
<OVSSwitch s23:
lo:127.0.0.1,s23-eth1:None,s23-eth2:None,s23-eth3:None,s23-eth4:None,s23-eth5:None,s2
3-eth6:None pid=12721>
<OVSSwitch s24:
lo:127.0.0.1,s24-eth1:None,s24-eth2:None,s24-eth3:None,s24-eth4:None,s24-eth5:None,s2
4-eth6:None pid=12724>

```
<OVSSwitch s25:
lo:127.0.0.1,s25-eth1:None,s25-eth2:None,s25-eth3:None,s25-eth4:None,s25-eth5:None,s2
5-eth6:None pid=12727>
<OVSSwitch s26:
lo:127.0.0.1,s26-eth1:None,s26-eth2:None,s26-eth3:None,s26-eth4:None,s26-eth5:None,s2
6-eth6:None pid=12730>
<OVSSwitch s27:
lo:127.0.0.1,s27-eth1:None,s27-eth2:None,s27-eth3:None,s27-eth4:None,s27-eth5:None,s2
7-eth6:None pid=12733>
<OVSSwitch s28:
lo:127.0.0.1,s28-eth1:None,s28-eth2:None,s28-eth3:None,s28-eth4:None,s28-eth5:None,s2
8-eth6:None pid=12736>
<OVSSwitch s29:
lo:127.0.0.1,s29-eth1:None,s29-eth2:None,s29-eth3:None,s29-eth4:None,s29-eth5:None,s2
9-eth6:None pid=12739>
<OVSSwitch s30:
lo:127.0.0.1,s30-eth1:None,s30-eth2:None,s30-eth3:None,s30-eth4:None,s30-eth5:None,s3
0-eth6:None pid=12742>
<OVSSwitch s31:
lo:127.0.0.1,s31-eth1:None,s31-eth2:None,s31-eth3:None,s31-eth4:None,s31-eth5:None,s3
1-eth6:None pid=12745>
<Controller c0: 127.0.0.1:6653 pid=12393>
```

mininet> h1 ifconfig -a

```
h1-eth0  Link encap:Ethernet  HWaddr 00:00:00:00:00:01
          inet addr:10.0.0.1  Bcast:10.255.255.255  Mask:255.0.0.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

```
lo       Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

mininet> s1 ifconfig -a

```
eth0     Link encap:Ethernet  HWaddr 08:00:27:7e:0f:24
          inet addr:192.168.56.104  Bcast:192.168.56.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:17890 errors:0 dropped:0 overruns:0 frame:0
          TX packets:26899 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1746034 (1.7 MB)  TX bytes:3878017 (3.8 MB)
```

eth1 Link encap:Ethernet HWaddr 08:00:27:a7:c9:40
inet addr:10.0.2.15 Bcast:10.0.2.255 Mask:255.255.255.0
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:559 errors:0 dropped:0 overruns:0 frame:0
TX packets:589 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:50879 (50.8 KB) TX bytes:51175 (51.1 KB)

lo Link encap:Local Loopback
inet addr:127.0.0.1 Mask:255.0.0.0
UP LOOPBACK RUNNING MTU:65536 Metric:1
RX packets:1479048 errors:0 dropped:0 overruns:0 frame:0
TX packets:1479048 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:152123562 (152.1 MB) TX bytes:152123562 (152.1 MB)

ovs-system Link encap:Ethernet HWaddr 0e:32:8f:9b:83:34
BROADCAST MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s1 Link encap:Ethernet HWaddr 6a:07:f1:87:14:4b
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s2 Link encap:Ethernet HWaddr e6:fc:b7:5d:28:4a
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s3 Link encap:Ethernet HWaddr f2:ca:6f:5a:62:4e
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s4 Link encap:Ethernet HWaddr a2:8a:4c:53:d3:47
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0

collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s5 Link encap:Ethernet HWaddr 36:6b:b5:81:85:4c
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s6 Link encap:Ethernet HWaddr 86:d8:d7:57:f5:4e
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s7 Link encap:Ethernet HWaddr 8e:27:f6:06:b7:47
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s8 Link encap:Ethernet HWaddr 32:ab:3c:99:03:46
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s9 Link encap:Ethernet HWaddr c6:b3:c4:c2:0d:4d
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s10 Link encap:Ethernet HWaddr ea:c7:22:32:24:48
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s11 Link encap:Ethernet HWaddr f2:3a:be:e6:e3:49
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0

TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s12 Link encap:Ethernet HWaddr c6:b2:1b:02:d6:46
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s13 Link encap:Ethernet HWaddr 0a:0f:1c:e0:ea:41
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s14 Link encap:Ethernet HWaddr 6a:d3:0a:e2:d3:44
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s15 Link encap:Ethernet HWaddr 56:96:fc:a8:65:4f
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s16 Link encap:Ethernet HWaddr ca:77:4b:e6:9e:4b
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s17 Link encap:Ethernet HWaddr d2:e0:42:e5:37:4c
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s18 Link encap:Ethernet HWaddr 2a:03:69:e0:c7:41
UP BROADCAST RUNNING MTU:1500 Metric:1

RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s19 Link encap:Ethernet HWaddr d6:96:3d:9f:a8:40
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s20 Link encap:Ethernet HWaddr f2:13:1c:24:4b:49
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s21 Link encap:Ethernet HWaddr 06:04:d9:3b:e9:4e
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s22 Link encap:Ethernet HWaddr 4e:6d:46:48:4a:4b
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s23 Link encap:Ethernet HWaddr e6:51:97:38:16:4a
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s24 Link encap:Ethernet HWaddr 62:68:bb:d8:6b:41
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s25 Link encap:Ethernet HWaddr 52:45:f8:2c:be:49

UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s26 Link encap:Ethernet HWaddr f6:f4:5f:f7:c1:42
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s27 Link encap:Ethernet HWaddr f2:70:6e:0c:98:49
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s28 Link encap:Ethernet HWaddr ce:a6:ea:4c:0e:47
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s29 Link encap:Ethernet HWaddr 52:21:43:de:e6:46
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s30 Link encap:Ethernet HWaddr d6:cb:f1:b9:ee:4e
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s31 Link encap:Ethernet HWaddr 92:84:ee:84:40:43
UP BROADCAST RUNNING MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s1-eth1 Link encap:Ethernet HWaddr 96:0e:d3:b4:77:f0
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s1-eth2 Link encap:Ethernet HWaddr 2a:31:d3:99:ed:c3
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s1-eth3 Link encap:Ethernet HWaddr 86:bd:a0:c6:7e:01
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s1-eth4 Link encap:Ethernet HWaddr 5e:79:94:a9:a7:09
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s1-eth5 Link encap:Ethernet HWaddr 92:10:f5:8b:db:ce
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s10-eth1 Link encap:Ethernet HWaddr 4e:e4:32:6e:bc:57
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s10-eth2 Link encap:Ethernet HWaddr c6:2c:e2:f9:32:0d
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s10-eth3 Link encap:Ethernet HWaddr fe:e5:14:51:77:2c
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s10-eth4 Link encap:Ethernet HWaddr 42:d0:0c:9d:24:c1
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s10-eth5 Link encap:Ethernet HWaddr e6:28:fe:10:fd:72
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s10-eth6 Link encap:Ethernet HWaddr 3a:28:97:9a:06:64
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s11-eth1 Link encap:Ethernet HWaddr 3e:8e:12:04:4a:7e
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s11-eth2 Link encap:Ethernet HWaddr 9a:1b:5d:8e:6c:9a
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s11-eth3 Link encap:Ethernet HWaddr 16:3b:e7:54:5c:04
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000

RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s11-eth4 Link encap:Ethernet HWaddr 7e:d0:95:65:29:73
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s11-eth5 Link encap:Ethernet HWaddr f6:e3:f4:91:4b:92
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s11-eth6 Link encap:Ethernet HWaddr ea:7d:f0:36:8f:bf
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s12-eth1 Link encap:Ethernet HWaddr ce:db:af:c3:ac:80
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s12-eth2 Link encap:Ethernet HWaddr ce:89:27:f4:eb:73
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s12-eth3 Link encap:Ethernet HWaddr 96:df:f0:29:e3:1a
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s12-eth4 Link encap:Ethernet HWaddr 66:e3:99:ee:9b:05
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0

collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s12-eth5 Link encap:Ethernet HWaddr 72:ed:3c:a7:fd:a4
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s12-eth6 Link encap:Ethernet HWaddr 6a:d1:da:3d:3d:c7
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s13-eth1 Link encap:Ethernet HWaddr 3a:22:38:e7:4a:0b
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s13-eth2 Link encap:Ethernet HWaddr 3e:72:33:db:98:46
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s13-eth3 Link encap:Ethernet HWaddr 5a:c0:db:f2:1d:fc
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s13-eth4 Link encap:Ethernet HWaddr a2:5f:64:e5:bf:47
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s13-eth5 Link encap:Ethernet HWaddr 52:86:d8:66:d4:a2
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0

TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s13-eth6 Link encap:Ethernet HWaddr f2:1d:60:3d:58:b3
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s14-eth1 Link encap:Ethernet HWaddr e2:bf:e4:98:02:81
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s14-eth2 Link encap:Ethernet HWaddr f6:a0:17:46:8e:03
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s14-eth3 Link encap:Ethernet HWaddr 0a:70:51:c9:73:98
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s14-eth4 Link encap:Ethernet HWaddr 8e:0e:c3:c5:28:ac
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s14-eth5 Link encap:Ethernet HWaddr 0a:2b:da:5b:b1:52
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s14-eth6 Link encap:Ethernet HWaddr 5e:92:1a:91:e8:47
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1

RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s15-eth1 Link encap:Ethernet HWaddr 42:f3:f3:08:29:88
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s15-eth2 Link encap:Ethernet HWaddr fe:30:d0:68:12:43
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s15-eth3 Link encap:Ethernet HWaddr 42:e3:d2:3b:7e:b4
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s15-eth4 Link encap:Ethernet HWaddr 12:b0:90:5a:af:16
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s15-eth5 Link encap:Ethernet HWaddr 86:8c:33:42:fa:73
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s15-eth6 Link encap:Ethernet HWaddr 92:e3:60:bd:92:57
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s16-eth1 Link encap:Ethernet HWaddr 3e:23:99:e8:52:d6

UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s16-eth2 Link encap:Ethernet HWaddr 7a:0d:d9:03:93:92
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s16-eth3 Link encap:Ethernet HWaddr 2e:67:e7:e4:32:4f
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s16-eth4 Link encap:Ethernet HWaddr 9e:01:26:8a:00:8b
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s16-eth5 Link encap:Ethernet HWaddr 7a:b8:9d:75:e0:cc
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s16-eth6 Link encap:Ethernet HWaddr 0a:e8:05:34:ab:4e
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s17-eth1 Link encap:Ethernet HWaddr ea:b4:77:42:39:56
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s17-eth2 Link encap:Ethernet HWaddr de:28:eb:82:73:dc
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s17-eth3 Link encap:Ethernet HWaddr 82:65:9c:19:da:0d
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s17-eth4 Link encap:Ethernet HWaddr 46:32:60:85:03:11
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s17-eth5 Link encap:Ethernet HWaddr ce:c9:ff:60:c2:75
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s17-eth6 Link encap:Ethernet HWaddr 6a:43:57:b7:ee:f4
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s18-eth1 Link encap:Ethernet HWaddr 76:42:80:5d:bb:32
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s18-eth2 Link encap:Ethernet HWaddr a6:e4:ca:e4:3c:bc
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s18-eth3 Link encap:Ethernet HWaddr d6:c2:29:5f:fe:d5
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s18-eth4 Link encap:Ethernet HWaddr 66:3b:25:f3:3d:6e
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s18-eth5 Link encap:Ethernet HWaddr e2:70:6a:57:d0:4b
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s18-eth6 Link encap:Ethernet HWaddr ee:2e:42:8f:6c:b4
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s19-eth1 Link encap:Ethernet HWaddr c6:e4:21:65:81:dd
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s19-eth2 Link encap:Ethernet HWaddr c6:23:b4:25:2a:f6
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s19-eth3 Link encap:Ethernet HWaddr 2a:24:49:0b:2d:02
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000

RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s19-eth4 Link encap:Ethernet HWaddr 3e:fd:7e:f5:aa:4b
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s19-eth5 Link encap:Ethernet HWaddr c6:84:88:ec:87:94
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s19-eth6 Link encap:Ethernet HWaddr 26:88:8d:d0:90:6a
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s2-eth1 Link encap:Ethernet HWaddr e2:b1:0e:8c:2c:ff
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s2-eth2 Link encap:Ethernet HWaddr ea:63:c8:9e:49:38
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s2-eth3 Link encap:Ethernet HWaddr 42:d4:87:3f:b6:43
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s2-eth4 Link encap:Ethernet HWaddr c6:97:e9:b1:b4:9c
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0

collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s2-eth5 Link encap:Ethernet HWaddr 02:f0:0f:03:dc:00
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s2-eth6 Link encap:Ethernet HWaddr ba:1e:43:af:37:3c
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s20-eth1 Link encap:Ethernet HWaddr c2:4d:f9:2d:7e:58
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s20-eth2 Link encap:Ethernet HWaddr 1e:92:5a:53:2e:3f
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s20-eth3 Link encap:Ethernet HWaddr 9e:90:a1:00:50:0a
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s20-eth4 Link encap:Ethernet HWaddr a6:a7:78:dc:94:33
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s20-eth5 Link encap:Ethernet HWaddr de:ed:39:eb:8e:f3
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0

TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s20-eth6 Link encap:Ethernet HWaddr 72:41:87:0e:bd:27
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s21-eth1 Link encap:Ethernet HWaddr 46:4e:20:aa:6c:fa
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s21-eth2 Link encap:Ethernet HWaddr 7e:20:41:d2:10:8d
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s21-eth3 Link encap:Ethernet HWaddr 12:cf:9f:80:e2:41
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s21-eth4 Link encap:Ethernet HWaddr ca:63:9c:c9:6d:3f
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s21-eth5 Link encap:Ethernet HWaddr 02:66:a8:a5:a5:14
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s21-eth6 Link encap:Ethernet HWaddr 26:78:8a:00:35:19
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1

RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s22-eth1 Link encap:Ethernet HWaddr de:e5:23:38:5d:e9
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s22-eth2 Link encap:Ethernet HWaddr ae:ae:c0:86:0f:d8
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s22-eth3 Link encap:Ethernet HWaddr ee:0e:fb:d4:11:5a
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s22-eth4 Link encap:Ethernet HWaddr a2:3e:93:da:0f:bf
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s22-eth5 Link encap:Ethernet HWaddr aa:14:d5:28:c4:8e
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s22-eth6 Link encap:Ethernet HWaddr 9a:97:66:b2:80:07
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s23-eth1 Link encap:Ethernet HWaddr b2:e6:f2:ed:ec:0e

UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s23-eth2 Link encap:Ethernet HWaddr 36:4a:a8:e0:8b:c0
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s23-eth3 Link encap:Ethernet HWaddr 42:30:a9:8a:57:02
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s23-eth4 Link encap:Ethernet HWaddr 5a:27:21:c0:d0:9d
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s23-eth5 Link encap:Ethernet HWaddr 9e:b0:cd:85:75:58
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s23-eth6 Link encap:Ethernet HWaddr ee:3a:09:a9:c2:49
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s24-eth1 Link encap:Ethernet HWaddr 72:5f:15:ac:86:f5
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s24-eth2 Link encap:Ethernet HWaddr aa:14:d2:0e:1e:e6
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s24-eth3 Link encap:Ethernet HWaddr ee:da:18:4b:7c:00
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s24-eth4 Link encap:Ethernet HWaddr ce:a4:e7:5a:e7:1d
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s24-eth5 Link encap:Ethernet HWaddr de:4a:26:ed:b5:d0
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s24-eth6 Link encap:Ethernet HWaddr ce:7d:b9:8a:7b:37
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s25-eth1 Link encap:Ethernet HWaddr 8a:57:ce:77:cd:f8
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s25-eth2 Link encap:Ethernet HWaddr 4a:4e:93:56:4c:0e
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s25-eth3 Link encap:Ethernet HWaddr 62:57:e4:47:9f:bb
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s25-eth4 Link encap:Ethernet HWaddr 8e:c2:63:ee:96:49
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s25-eth5 Link encap:Ethernet HWaddr ca:36:71:5f:ec:68
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s25-eth6 Link encap:Ethernet HWaddr 9a:97:5f:e1:1c:81
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s26-eth1 Link encap:Ethernet HWaddr 02:9e:d7:be:12:11
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s26-eth2 Link encap:Ethernet HWaddr fe:02:b4:a4:7b:eb
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s26-eth3 Link encap:Ethernet HWaddr 42:23:e6:38:a6:c3
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000

RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s26-eth4 Link encap:Ethernet HWaddr 92:5b:e9:93:64:d7
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s26-eth5 Link encap:Ethernet HWaddr de:64:76:c4:d6:8e
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s26-eth6 Link encap:Ethernet HWaddr 4a:99:c6:76:f1:a5
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s27-eth1 Link encap:Ethernet HWaddr 16:93:3b:17:ef:f1
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s27-eth2 Link encap:Ethernet HWaddr 3e:9b:ae:53:39:44
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s27-eth3 Link encap:Ethernet HWaddr e6:6f:0b:f0:26:42
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s27-eth4 Link encap:Ethernet HWaddr 7e:2d:38:00:ee:b3
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0

collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s27-eth5 Link encap:Ethernet HWaddr b6:cc:0c:ef:d9:c7
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s27-eth6 Link encap:Ethernet HWaddr 42:f8:1d:dc:56:23
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s28-eth1 Link encap:Ethernet HWaddr de:1f:eb:f6:2b:86
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s28-eth2 Link encap:Ethernet HWaddr 96:ad:51:7a:73:50
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s28-eth3 Link encap:Ethernet HWaddr 8a:9d:18:af:da:47
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s28-eth4 Link encap:Ethernet HWaddr 4a:af:08:2e:5b:6b
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s28-eth5 Link encap:Ethernet HWaddr 62:84:e8:bc:f6:12
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0

TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s28-eth6 Link encap:Ethernet HWaddr 12:b6:79:9f:cf:d9
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s29-eth1 Link encap:Ethernet HWaddr ea:5a:64:49:a3:8c
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s29-eth2 Link encap:Ethernet HWaddr 76:fc:82:f8:1e:aa
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s29-eth3 Link encap:Ethernet HWaddr d2:41:02:f2:a1:9b
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s29-eth4 Link encap:Ethernet HWaddr b2:8c:88:c0:e8:fa
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s29-eth5 Link encap:Ethernet HWaddr d2:43:79:15:5c:14
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s29-eth6 Link encap:Ethernet HWaddr c2:98:60:02:ae:89
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1

RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s3-eth1 Link encap:Ethernet HWaddr 16:2b:0b:37:e6:ae
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s3-eth2 Link encap:Ethernet HWaddr de:f1:2e:bb:74:7f
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s3-eth3 Link encap:Ethernet HWaddr c2:bb:96:e1:4a:49
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s3-eth4 Link encap:Ethernet HWaddr 22:6b:12:e5:e8:0f
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s3-eth5 Link encap:Ethernet HWaddr da:5f:d9:43:28:f9
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s3-eth6 Link encap:Ethernet HWaddr 36:1d:ad:4a:ee:d5
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s30-eth1 Link encap:Ethernet HWaddr e6:73:25:61:cf:8c

UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s30-eth2 Link encap:Ethernet HWaddr ee:b6:7b:6d:3e:bf
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s30-eth3 Link encap:Ethernet HWaddr 56:42:1f:50:a6:ea
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s30-eth4 Link encap:Ethernet HWaddr 72:43:d3:24:c9:57
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s30-eth5 Link encap:Ethernet HWaddr 42:61:6e:41:c0:ed
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s30-eth6 Link encap:Ethernet HWaddr 9e:fa:37:b2:b5:32
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s31-eth1 Link encap:Ethernet HWaddr 82:66:63:8d:1d:7b
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s31-eth2 Link encap:Ethernet HWaddr a6:cb:48:6d:61:de
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s31-eth3 Link encap:Ethernet HWaddr 62:0c:71:af:45:0a
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s31-eth4 Link encap:Ethernet HWaddr 32:0b:d7:77:71:63
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s31-eth5 Link encap:Ethernet HWaddr 62:44:cc:5c:d1:3c
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s31-eth6 Link encap:Ethernet HWaddr e2:4b:d6:d4:c2:3e
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s4-eth1 Link encap:Ethernet HWaddr 92:5d:9f:6d:aa:9f
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s4-eth2 Link encap:Ethernet HWaddr 6e:a4:dd:b3:bf:3d
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s4-eth3 Link encap:Ethernet HWaddr 6a:c2:52:f5:c4:de
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s4-eth4 Link encap:Ethernet HWaddr 96:5d:f8:3a:3e:53
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s4-eth5 Link encap:Ethernet HWaddr 62:11:7a:40:a4:25
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s4-eth6 Link encap:Ethernet HWaddr 5a:66:c9:83:92:48
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s5-eth1 Link encap:Ethernet HWaddr f2:22:26:a2:0c:29
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s5-eth2 Link encap:Ethernet HWaddr de:8e:25:fd:e6:8f
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s5-eth3 Link encap:Ethernet HWaddr 4a:5d:bb:1b:08:7a
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000

RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s5-eth4 Link encap:Ethernet HWaddr 82:97:65:21:f6:96
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s5-eth5 Link encap:Ethernet HWaddr 42:2d:c7:46:c9:e3
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s5-eth6 Link encap:Ethernet HWaddr 56:d5:ea:40:a8:68
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s6-eth1 Link encap:Ethernet HWaddr 3a:61:2b:5c:67:9f
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s6-eth2 Link encap:Ethernet HWaddr 5e:27:db:46:cd:ca
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s6-eth3 Link encap:Ethernet HWaddr 22:0a:cf:07:d3:14
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s6-eth4 Link encap:Ethernet HWaddr f6:63:92:95:66:6f
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0

collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s6-eth5 Link encap:Ethernet HWaddr 02:82:60:5f:f7:69
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s6-eth6 Link encap:Ethernet HWaddr 7a:8c:6a:bf:3f:cc
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s7-eth1 Link encap:Ethernet HWaddr 12:fc:1e:af:2a:51
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s7-eth2 Link encap:Ethernet HWaddr e2:f5:88:af:f0:92
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s7-eth3 Link encap:Ethernet HWaddr 8a:57:a5:35:5d:e9
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s7-eth4 Link encap:Ethernet HWaddr 0a:ae:91:75:81:b4
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s7-eth5 Link encap:Ethernet HWaddr 0a:7c:dc:12:5b:5a
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0

TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s7-eth6 Link encap:Ethernet HWaddr de:44:30:e3:df:8b
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s8-eth1 Link encap:Ethernet HWaddr d6:64:e6:d6:f2:c0
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s8-eth2 Link encap:Ethernet HWaddr 9a:2d:95:4e:27:74
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s8-eth3 Link encap:Ethernet HWaddr 16:89:f5:d5:fc:f8
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s8-eth4 Link encap:Ethernet HWaddr 1a:78:b2:12:94:30
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s8-eth5 Link encap:Ethernet HWaddr 02:8a:0c:20:74:26
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s8-eth6 Link encap:Ethernet HWaddr 92:d3:c0:2a:8b:60
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1

RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s9-eth1 Link encap:Ethernet HWaddr 86:27:49:b8:e0:01
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s9-eth2 Link encap:Ethernet HWaddr 0a:7e:b7:ae:28:51
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s9-eth3 Link encap:Ethernet HWaddr e2:1c:9c:9a:c3:a6
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s9-eth4 Link encap:Ethernet HWaddr 66:3f:4f:e7:40:87
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s9-eth5 Link encap:Ethernet HWaddr 2a:89:d9:47:21:a4
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

s9-eth6 Link encap:Ethernet HWaddr 7a:cb:c7:7f:c9:f3
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

mininet> pingall

[illegible]

h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117
h118 h119 h120 h121 h122 h123 h124 h125

2 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101
h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117
h118 h119 h120 h121 h122 h123 h124 h125

2 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h61 h6
2 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h8
2 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101
h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117
h118 h119 h120 h121 h122 h123 h124 h125

h30 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h2
1 h22 h23 h24 h25 h26 h27 h28 h29 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h41 h4
2 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h61 h6
2 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h8
2 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101
h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117
h118 h119 h120 h121 h122 h123 h124 h125

h32 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h2
1 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h33 h34 h35 h36 h37 h38 h39 h40 h41 h4
2 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h61 h6
2 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h8
2 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101
h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117
h118 h119 h120 h121 h122 h123 h124 h125

h34 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h21 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h35 h36 h37 h38 h39 h40 h41 h42 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h61 h62 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h82 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101 h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117 h118 h119 h120 h121 h122 h123 h124 h125

h35 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h2
1 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h36 h37 h38 h39 h40 h41 h4
2 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h61 h6
2 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h8
2 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101
h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117
h118 h119 h120 h121 h122 h123 h124 h125

h37 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h2
1 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h38 h39 h40 h41 h4
2 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h61 h6
2 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h8
2 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101
h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117
h118 h119 h120 h121 h122 h123 h124 h125

h39 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h2
1 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h40 h41 h4
2 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h61 h6
2 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h8
2 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101
h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117
h118 h119 h120 h121 h122 h123 h124 h125

h41 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h2
1 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h4
2 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h61 h6
2 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h8
2 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101

2 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101
h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117
h118 h119 h120 h121 h122 h123 h124 h125

h50 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h2
1 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h4
1 h42 h43 h44 h45 h46 h47 h48 h49 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h61 h6
2 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h8
2 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101
h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117
h118 h119 h120 h121 h122 h123 h124 h125

h52 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h2
1 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h4
1 h42 h43 h44 h45 h46 h47 h48 h49 h50 h51 h53 h54 h55 h56 h57 h58 h59 h60 h61 h6
2 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h8
2 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101
h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117
h118 h119 h120 h121 h122 h123 h124 h125

h54 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h21 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h41 h42 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h55 h56 h57 h58 h59 h60 h61 h62 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h82 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101 h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117 h118 h119 h120 h121 h122 h123 h124 h125

1 h42 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h6
1 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h8
2 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101
h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117
h118 h119 h120 h121 h122 h123 h124 h125

h64 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h2
1 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h4
1 h42 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h6
1 h62 h63 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h8
2 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101
h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117
h118 h119 h120 h121 h122 h123 h124 h125

h65 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h21 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h41 h42 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h61 h62 h63 h64 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h82 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101 h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117 h118 h119 h120 h121 h122 h123 h124 h125

h66 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h21 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h41 h42 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h61 h62 h63 h64 h65 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h82 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101 h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117 h118 h119 h120 h121 h122 h123 h124 h125

h67 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h2
1 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h4
1 h42 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h61
h62 h63 h64 h65 h66 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h82 h83
h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101 h102 h103
h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117 h118 h119
h120 h121 h122 h123 h124 h125

h68 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h21 h22
h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h41 h42 h43
h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h61 h62 h63 h64
h65 h66 h67 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h82 h83 h84 h85 h86
h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101 h102 h103 h104 h105
h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117 h118 h119 h120 h121
h122 h123 h124 h125


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h75 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h2
1 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h4
1 h42 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h6
1 h62 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h76 h77 h78 h79 h80 h81 h8
2 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101
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h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117
h118 h119 h120 h121 h122 h123 h124 h125

1 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101
h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117
h118 h119 h120 h121 h122 h123 h124 h125

h84 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h2
1 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h4
1 h42 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h6
1 h62 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h8
1 h82 h83 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101
h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117
h118 h119 h120 h121 h122 h123 h124 h125

h86 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h2
1 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h4
1 h42 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h6
1 h62 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h8
1 h82 h83 h84 h85 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101
h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117
h118 h119 h120 h121 h122 h123 h124 h125

h88 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h21 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h41 h42 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h61 h62 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h82 h83 h84 h85 h86 h87 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101 h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117 h118 h119 h120 h121 h122 h123 h124 h125

1 h62 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81
1 h82 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101
h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117
h118 h119 h120 h121 h122 h123 h124 h125

h91 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h21 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h41 h42 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h61 h62 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h82 h83 h84 h85 h86 h87 h88 h89 h90 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101 h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117 h118 h119 h120 h121 h122 h123 h124 h125

h93 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h21 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h41 h42 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h61 h62 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h82 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h94 h95 h96 h97 h98 h99 h100 h101 h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117 h118 h119 h120 h121 h122 h123 h124 h125

h95 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h21 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h41 h42 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h61 h62 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h82 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h96 h97 h98 h99 h100 h101 h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117 h118 h119 h120 h121 h122 h123 h124 h125

1 h42 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h6
1 h62 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h8
1 h82 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h97 h98 h99 h100 h101
h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117
h118 h119 h120 h121 h122 h123 h124 h125

h98 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h21 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h41 h42 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h61 h62 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h82 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h99 h100 h101 h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117 h118 h119 h120 h121 h122 h123 h124 h125

h100 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h
21 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h
41 h42 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h
61 h62 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h
81 h82 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h101
h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117
h118 h119 h120 h121 h122 h123 h124 h125

h102 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h
21 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h
41 h42 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h
61 h62 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h
81 h82 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100
h101 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117
h118 h119 h120 h121 h122 h123 h124 h125

h103 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h
21 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h
41 h42 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h
61 h62 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h
81 h82 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100
h101 h102 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117
h118 h119 h120 h121 h122 h123 h124 h125

61 h62 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h
81 h82 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100
h101 h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116
h117 h118 h119 h120 h121 h122 h124 h125

h124 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h
21 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h
41 h42 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h
61 h62 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h
81 h82 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100
h101 h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116
h117 h118 h119 h120 h121 h122 h123 h125

h125 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h
21 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h
41 h42 h43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h
61 h62 h63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h
81 h82 h83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100
h101 h102 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116
h117 h118 h119 h120 h121 h122 h123 h124

*** Results: 0% dropped (15500/15500 received)

mininet> h1 ping -c 4 h2 & h1 ping -c 4 h125

64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=13.8 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.484 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.440 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.101 ms

--- 10.0.0.2 ping statistics ---

4 packets transmitted, 4 received, 0% packet loss, time 3001ms
rtt min/avg/max/mdev = 0.101/3.719/13.853/5.852 ms

mininet> xterm h1 h2

mininet> exit

*** Stopping 1 controllers

c0

*** Stopping 8 terms

*** Stopping 155 links

.....

.....

*** Stopping 31 switches

s1 s2 s3 s4 s5 s6 s7 s8 s9 s10 s11 s12 s13 s14 s15 s16 s17 s18 s19 s20 s21 s22 s
23 s24 s25 s26 s27 s28 s29 s30 s31

*** Stopping 125 hosts

h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h21 h22 h
23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h41 h42 h
43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h61 h62 h
63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h82 h
83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101 h10
2 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117 h11
8 h119 h120 h121 h122 h123 h124 h125

*** Done

completed in 2088.956 seconds

mininet@mininet-vm:~\$ sudo mn -c

*** Removing excess controllers/ofprotocols/ofdatapaths/pings/noxes

killall controller ofprotocol ofdatapath ping nox_core lt-nox_core ovs-openflowd

ovs-controller udpbwtest mnexec ivs 2> /dev/null

killall -9 controller ofprotocol ofdatapath ping nox_core lt-nox_core ovs-openfl

owd ovs-controller udpbwtest mnexec ivs 2> /dev/null

pkill -9 -f "sudo mnexec"

*** Removing junk from /tmp

rm -f /tmp/vconn* /tmp/vlogs* /tmp/*.out /tmp/*.log

*** Removing old X11 tunnels

*** Removing excess kernel datapaths

ps ax | egrep -o 'dp[0-9]+' | sed 's/dp/nl:/'

*** Removing OVS datapaths

ovs-vsctl --timeout=1 list-br

ovs-vsctl --timeout=1 list-br

*** Removing all links of the pattern foo-ethX

ip link show | egrep -o '([_[:alnum:]]+-eth[[:digit:]]+)'

ip link show

*** Killing stale mininet node processes

pkill -9 -f mininet:

*** Shutting down stale tunnels

pkill -9 -f Tunnel=Ethernet

pkill -9 -f .ssh/mn

rm -f ~/.ssh/mn/*

*** Cleanup complete.

mininet@mininet-vm:~\$ sudo mn --topo=tree,depth=3,fanout=5 --mac --link=tc,bw=40

*** Creating network

*** Adding controller

*** Adding hosts:

h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h21 h22 h
23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h41 h42 h
43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h61 h62 h
63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h82 h
83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101 h10
2 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117 h11
8 h119 h120 h121 h122 h123 h124 h125

*** Adding switches:

s1 s2 s3 s4 s5 s6 s7 s8 s9 s10 s11 s12 s13 s14 s15 s16 s17 s18 s19 s20 s21 s22 s
23 s24 s25 s26 s27 s28 s29 s30 s31

*** Adding links:

(40.00Mbit) (40.00Mbit) (s1, s2) (40.00Mbit) (40.00Mbit) (s1, s8) (40.00Mbit) (4
0.00Mbit) (s1, s14) (40.00Mbit) (40.00Mbit) (s1, s20) (40.00Mbit) (40.00Mbit) (s
1, s26) (40.00Mbit) (40.00Mbit) (s2, s3) (40.00Mbit) (40.00Mbit) (s2, s4) (40.00
Mbit) (40.00Mbit) (s2, s5) (40.00Mbit) (40.00Mbit) (s2, s6) (40.00Mbit) (40.00Mb
it) (s2, s7) (40.00Mbit) (40.00Mbit) (s3, h1) (40.00Mbit) (40.00Mbit) (s3, h2) (
40.00Mbit) (40.00Mbit) (s3, h3) (40.00Mbit) (40.00Mbit) (s3, h4) (40.00Mbit) (40
.00Mbit) (s3, h5) (40.00Mbit) (40.00Mbit) (s4, h6) (40.00Mbit) (40.00Mbit) (s4,

h7) (40.00Mbit) (40.00Mbit) (s4, h8) (40.00Mbit) (40.00Mbit) (s4, h9) (40.00Mbit)
) (40.00Mbit) (s4, h10) (40.00Mbit) (40.00Mbit) (s5, h11) (40.00Mbit) (40.00Mbit)
) (s5, h12) (40.00Mbit) (40.00Mbit) (s5, h13) (40.00Mbit) (40.00Mbit) (s5, h14)
 (40.00Mbit) (40.00Mbit) (s5, h15) (40.00Mbit) (40.00Mbit) (s6, h16) (40.00Mbit)
 (40.00Mbit) (s6, h17) (40.00Mbit) (40.00Mbit) (s6, h18) (40.00Mbit) (40.00Mbit)
 (s6, h19) (40.00Mbit) (40.00Mbit) (s6, h20) (40.00Mbit) (40.00Mbit) (s7, h21) (4
 0.00Mbit) (40.00Mbit) (s7, h22) (40.00Mbit) (40.00Mbit) (s7, h23) (40.00Mbit) (4
 0.00Mbit) (s7, h24) (40.00Mbit) (40.00Mbit) (s7, h25) (40.00Mbit) (40.00Mbit) (s
 8, s9) (40.00Mbit) (40.00Mbit) (s8, s10) (40.00Mbit) (40.00Mbit) (s8, s11) (40.0
 0Mbit) (40.00Mbit) (s8, s12) (40.00Mbit) (40.00Mbit) (s8, s13) (40.00Mbit) (40.0
 0Mbit) (s9, h26) (40.00Mbit) (40.00Mbit) (s9, h27) (40.00Mbit) (40.00Mbit) (s9,
 h28) (40.00Mbit) (40.00Mbit) (s9, h29) (40.00Mbit) (40.00Mbit) (s9, h30) (40.00M
 bit) (40.00Mbit) (s10, h31) (40.00Mbit) (40.00Mbit) (s10, h32) (40.00Mbit) (40.0
 0Mbit) (s10, h33) (40.00Mbit) (40.00Mbit) (s10, h34) (40.00Mbit) (40.00Mbit) (s1
 0, h35) (40.00Mbit) (40.00Mbit) (s11, h36) (40.00Mbit) (40.00Mbit) (s11, h37) (4
 0.00Mbit) (40.00Mbit) (s11, h38) (40.00Mbit) (40.00Mbit) (s11, h39) (40.00Mbit)
 (40.00Mbit) (s11, h40) (40.00Mbit) (40.00Mbit) (s12, h41) (40.00Mbit) (40.00Mbit
) (s12, h42) (40.00Mbit) (40.00Mbit) (s12, h43) (40.00Mbit) (40.00Mbit) (s12, h4
 4) (40.00Mbit) (40.00Mbit) (s12, h45) (40.00Mbit) (40.00Mbit) (s13, h46) (40.00M
 bit) (40.00Mbit) (s13, h47) (40.00Mbit) (40.00Mbit) (s13, h48) (40.00Mbit) (40.0
 0Mbit) (s13, h49) (40.00Mbit) (40.00Mbit) (s13, h50) (40.00Mbit) (40.00Mbit) (s1
 4, s15) (40.00Mbit) (40.00Mbit) (s14, s16) (40.00Mbit) (40.00Mbit) (s14, s17) (4
 0.00Mbit) (40.00Mbit) (s14, s18) (40.00Mbit) (40.00Mbit) (s14, s19) (40.00Mbit)
 (40.00Mbit) (s15, h51) (40.00Mbit) (40.00Mbit) (s15, h52) (40.00Mbit) (40.00Mbit
) (s15, h53) (40.00Mbit) (40.00Mbit) (s15, h54) (40.00Mbit) (40.00Mbit) (s15, h5
 5) (40.00Mbit) (40.00Mbit) (s16, h56) (40.00Mbit) (40.00Mbit) (s16, h57) (40.00M
 bit) (40.00Mbit) (s16, h58) (40.00Mbit) (40.00Mbit) (s16, h59) (40.00Mbit) (40.0
 0Mbit) (s16, h60) (40.00Mbit) (40.00Mbit) (s17, h61) (40.00Mbit) (40.00Mbit) (s1
 7, h62) (40.00Mbit) (40.00Mbit) (s17, h63) (40.00Mbit) (40.00Mbit) (s17, h64) (4
 0.00Mbit) (40.00Mbit) (s17, h65) (40.00Mbit) (40.00Mbit) (s18, h66) (40.00Mbit)
 (40.00Mbit) (s18, h67) (40.00Mbit) (40.00Mbit) (s18, h68) (40.00Mbit) (40.00Mbit
) (s18, h69) (40.00Mbit) (40.00Mbit) (s18, h70) (40.00Mbit) (40.00Mbit) (s19, h7
 1) (40.00Mbit) (40.00Mbit) (s19, h72) (40.00Mbit) (40.00Mbit) (s19, h73) (40.00M
 bit) (40.00Mbit) (s19, h74) (40.00Mbit) (40.00Mbit) (s19, h75) (40.00Mbit) (40.0
 0Mbit) (s20, s21) (40.00Mbit) (40.00Mbit) (s20, s22) (40.00Mbit) (40.00Mbit) (s2
 0, s23) (40.00Mbit) (40.00Mbit) (s20, s24) (40.00Mbit) (40.00Mbit) (s20, s25) (4
 0.00Mbit) (40.00Mbit) (s21, h76) (40.00Mbit) (40.00Mbit) (s21, h77) (40.00Mbit)
 (40.00Mbit) (s21, h78) (40.00Mbit) (40.00Mbit) (s21, h79) (40.00Mbit) (40.00Mbit
) (s21, h80) (40.00Mbit) (40.00Mbit) (s22, h81) (40.00Mbit) (40.00Mbit) (s22, h8
 2) (40.00Mbit) (40.00Mbit) (s22, h83) (40.00Mbit) (40.00Mbit) (s22, h84) (40.00M
 bit) (40.00Mbit) (s22, h85) (40.00Mbit) (40.00Mbit) (s23, h86) (40.00Mbit) (40.0
 0Mbit) (s23, h87) (40.00Mbit) (40.00Mbit) (s23, h88) (40.00Mbit) (40.00Mbit) (s2
 3, h89) (40.00Mbit) (40.00Mbit) (s23, h90) (40.00Mbit) (40.00Mbit) (s24, h91) (4
 0.00Mbit) (40.00Mbit) (s24, h92) (40.00Mbit) (40.00Mbit) (s24, h93) (40.00Mbit)
 (40.00Mbit) (s24, h94) (40.00Mbit) (40.00Mbit) (s24, h95) (40.00Mbit) (40.00Mbit
) (s25, h96) (40.00Mbit) (40.00Mbit) (s25, h97) (40.00Mbit) (40.00Mbit) (s25, h9
 8) (40.00Mbit) (40.00Mbit) (s25, h99) (40.00Mbit) (40.00Mbit) (s25, h100) (40.00
 Mbit) (40.00Mbit) (s26, s27) (40.00Mbit) (40.00Mbit) (s26, s28) (40.00Mbit) (40.

0.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit)
(40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00
Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit) (4
0.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit
) (40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00
Mbit) (40.00Mbit)

*** Starting CLI:

mininet> xterm h1 h2

mininet> exit

*** Stopping 1 controllers

c0

*** Stopping 4 terms

*** Stopping 155 links

.....
.....

*** Stopping 31 switches

sudos1 s2 s3 s4 s5 s6 s7 s8 s9 s10 s11 s12 s13 s14 s15 s16 s17 s18 s19 s20 s21 s
22 s23 s24 s25 s26 s27 s28 s29 s30 s31

*** Stopping 125 hosts

h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h21 h22 h
23 h24 h25 h26 h27 h28 h29 h30 h31 h32 h33 h34 h35 h36 h37 h38 h39 h40 h41 h42 h
43 h44 h45 h46 h47 h48 h49 h50 h51 h52 h53 h54 h55 h56 h57 h58 h59 h60 h61 h62 h
63 h64 h65 h66 h67 h68 h69 h70 h71 h72 h73 h74 h75 h76 h77 h78 h79 h80 h81 h82 h
83 h84 h85 h86 h87 h88 h89 h90 h91 h92 h93 h94 h95 h96 h97 h98 h99 h100 h101 h10
2 h103 h104 h105 h106 h107 h108 h109 h110 h111 h112 h113 h114 h115 h116 h117 h11
8 h119 h120 h121 h122 h123 h124 h125

*** Done

completed in 254.877 seconds

mininet@mininet-vm:~\$ sudo mn -c

*** Removing excess controllers/ofprotocols/ofdatapaths/pings/noxes

killall controller ofprotocol ofdatapath ping nox_core lt-nox_core ovs-openflowd
ovs-controller udpbwtest mnexec ivs 2> /dev/null

killall -9 controller ofprotocol ofdatapath ping nox_core lt-nox_core ovs-openfl
owd ovs-controller udpbwtest mnexec ivs 2> /dev/null

pkill -9 -f "sudo mnexec"

*** Removing junk from /tmp

rm -f /tmp/vconn* /tmp/vlogs* /tmp/*.out /tmp/*.log

*** Removing old X11 tunnels

*** Removing excess kernel datapaths

ps ax | egrep -o 'dp[0-9]+' | sed 's/dp/nl:/'

*** Removing OVS datapaths

ovs-vsctl --timeout=1 list-br

ovs-vsctl --timeout=1 list-br

*** Removing all links of the pattern foo-ethX

ip link show | egrep -o '([_[:alnum:]]+-eth[[:digit:]]+)'

ip link show

*** Killing stale mininet node processes

pkill -9 -f mininet:

*** Shutting down stale tunnels

pkill -9 -f Tunnel=Ethernet

pkill -9 -f .ssh/mn

rm -f ~/.ssh/mn/*

*** Cleanup complete.

mininet@mininet-vm:~\$